SAT Math Test VII — Easy to Hard

1 ★ Mark for Review ABC

A chemist mixed x liters of a 6% saline solution with y liters of a 9% saline solution to produce a 7% saline solution. Which equation best represents this situation? (Assume the volumes of the solutions are additive.)

A. $0.06x + 0.09y = 7(x + y)$

B. $0.06x + 0.09y = 0.07(x + y)$

C. $0.6x + 0.9y = 7(x + y)$

D. $0.6x + 0.9y = 0.7(x + y)$

2 ★ Mark for Review ABC

A window repair specialist charges \$220 for the first two hours of repair plus an hourly fee for each additional hour. The total cost for 5 hours of repair is \$400. Which function f gives the total cost, in dollars, for x hours of repair, where $x \geq 2$?

A. $f(x) = 60x + 100$

B. $f(x) = 60x + 220$

C. $f(x) = 80x$

D. $f(x) = 80x + 220$

3 ★ Mark for Review ABC

In the xy-plane, the graph of the linear function f has a negative slope and a y-intercept at $(0, k)$, where k is a positive constant. Which of the following must be true?

I. If $a > b$, then $f(a) < f(b)$.
II. If $a < 0$, then $f(a) > k$.
III. If $a > 0$, then $f(a) > 0$.

A. I and II only

B. I and III only

C. II and III only

D. I, II, and III

4 ★ Mark for Review ABC

In the given equation, k is a positive constant. What is the product of the solutions to the given equation?

$$x^2 - 8kx - (k - 4) = 0$$

A. $4 - k$

B. $k - 4$

C. $-8k$

D. $8k$

5 ★ Mark for Review ABC

The equation $2|x-5| = k$, where k is a constant, has exactly one solution. Which of the following could be the value of $\frac{k}{2}$?

A. -5 only

B. -5 or 5

C. 0 only

D. 5 only

6 ★ Mark for Review ABC

A manufacturer uses two assembly lines to produce refrigerator units. The table summarizes the results of a quality-control test used to check for defective units. If a refrigerator unit that was checked during this test is selected at random, what is the probability that the unit was produced by assembly line A, given that the unit is defective?

	Defective	Not defective	Total
Assembly line A	300	5700	6000
Assembly line B	500	3500	4000
Total	800	9200	10000

A. $\frac{1}{20}$

B. $\frac{3}{8}$

C. $\frac{3}{5}$

D. $\frac{5}{8}$

7 ★ Mark for Review ABC

A researcher wanted to determine whether studying at a library is more effective than studying at home. He selected a random sample of 200 students and gave them three days to study for the same test. The 100 students who lived closest to a library were asked to study at a library, and the remaining 100 students were asked to study at home. After the students took the test, the scores of those who studied at a library were compared to the scores of those who studied at home. How should this experiment be changed so that the researcher can conclude whether studying at a library is more effective than studying at home?

A. The students assigned to study at a library should study at the same library.

B. The students should be randomly assigned to study either at a library or at home.

C. The students assigned to study at a library should take a more difficult test than the students assigned to study at home.

D. No changes to the experiment are necessary.

8 ★ Mark for Review ABC

All the residents of a town were mailed a survey about proposed renovations to the town's high school. Approximately 10% of the residents completed the survey and mailed it back. Based on the responses, it was estimated that 72% of the residents support the proposed renovations, with an associated margin of error of 12%. Which of the following best explains why these survey results are unlikely to represent the opinions of all the town's residents?

A. The survey included residents who do not attend the high school.

B. Not all residents were aware of the proposed renovations before the survey.

C. Responses to the survey were not obtained from a random sample of the town's residents.

D. The margin of error is greater than the percentage of residents who responded to the survey.

9 ★ Mark for Review ABC

The estimated total pressure a diver experiences below the surface of the water is equal to the sum of the estimated water pressure and the estimated atmospheric pressure. At a depth of h meters below the surface, the estimated total pressure P_1, in kilopascals (kPa), is given by

$$P_1 = 10h + 101.$$

After maintaining a depth of h meters, the diver descended d additional meters. The estimated total pressure P_2, in kPa, after descending these d additional meters is given by

$$P_2 = 10d + 341.$$

What is the estimated total pressure, in kPa, the diver experienced at a depth of h meters below the surface?

A. 442

B. 341

C. 101

D. 10

10 ★ Mark for Review ABC

For each real number r, which of the following points lies on the graph of each equation in the xy-plane for the given system?

$$8x + 7y = 9$$
$$24x + 21y = 27$$

A. $\left(r, \dfrac{-8r+9}{7}\right)$

B. $\left(\dfrac{-8r+9}{7}, r\right)$

C. $\left(\dfrac{-8r+63}{7}, \dfrac{9r+189}{7}\right)$

D. $\left(\frac{r}{3}+9, -\frac{r}{3}+27\right)$

11 ★ Mark for Review ABC

An object's speed, in meters per hour, is increasing at a rate of 1,490 meters per hour per second. Which of the following is closest to this rate in feet per second squared? (Use 1 meter = 3.28 feet.)

A. 0.126

B. 1.36

C. 7.57

D. 81.5

12 ★ Mark for Review ABC

In the given system of equations, b is a constant. If the system has no solution, what is the value of b?

$$\frac{3}{10}y + \frac{b}{3}x = \frac{3}{4} - \frac{1}{5}y$$
$$\frac{3}{8}x + \frac{3}{5} = -\frac{1}{4}y + \frac{8}{5}$$

13 ★ Mark for Review ABC

A computer program models the population of a certain insect in an environment where the insect has no natural predators. According to the model, the estimated total mass of the population of insects at the end of every 5-week period is 158% greater than the estimated total mass at the end of the previous 5-week period. The estimated total mass at the end of 15 weeks is 627.7 grams. Which equation best represents this model, where M is the estimated total mass, in grams, at the end of t weeks?

A. $M = 538.92(1.58)^{t/5}$

B. $M = 457.66(2.58)^{t/5}$

C. $M = 159.14(1.58)^{t/5}$

D. $M = 36.55(2.58)^{t/5}$

14 ★ Mark for Review ABC

Which expression is a factor of

$$x^4 + 14ax^2 + 49a^2 - 64,$$

where a is a positive constant?

A. $x^2 + 7a + 8$

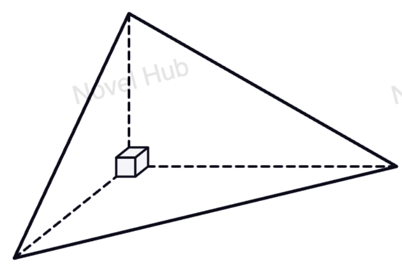
B. $x^2 - 7a - 8$

C. $x^2 + 14a$

D. $x^2 - 8a$

15 ★ Mark for Review ABC

The figure shown is a trirectangular tetrahedron, a three-dimensional figure with four triangular faces, three of which are right triangles. The base is an isosceles right triangle whose legs each have length 12 inches. The height of the tetrahedron is 24 inches. Which of the following is closest to the volume of this tetrahedron, in pints?



Note: Figure not drawn to scale.

Use 1 quart $= 57.75$ in^3, 1 pint $= 0.5$ quart, and $V = \frac{1}{3}Sh$, where S is the area of the base and h is the height.

A. 19.95

B. 119.69

C. 576.00

D. 16,632.00

16 ★ Mark for Review ABC

A right circular cone is inscribed in a sphere. The vertex of the cone is at the top of the sphere, and the center of the cone's circular base is 6 inches below the center of the sphere. The radius of the cone's base is 8 inches. What is the volume, in cubic inches, of the cone?

A. $\dfrac{512\pi}{3}$

B. $\dfrac{1024\pi}{3}$

C. 384π

D. $\frac{1280\pi}{3}$

17 ★ Mark for Review ABC

In the given quadratic function, b and c are constants. The graph of $y = f(x)$ in the xy-plane does not intersect the x-axis. If $f(-7) = f(9)$, which of the following could be the value of $b + c$?

$$f(x) = 3x^2 + bx + c$$

A. -1

B. -3

C. -5

D. -7

18 ★ Mark for Review ABC

In the given system of equations, p is a constant. The system has exactly one distinct real solution (x, y). What is the value of y?

$$y = 2x^2 + 15x + 57$$
$$y = -9x + p$$

19 ★ Mark for Review **ABC**

The graph of the quadratic function $y = |a|x^2 + bx + c$ passes through the following five points:

$$A(m, n), \quad B(0, y_1), \quad C(3 - m, n), \quad D(\sqrt{2}, y_2), \quad E(2, y_3).$$

What is the correct order of y_1, y_2, y_3?

A. $y_1 < y_2 < y_3$

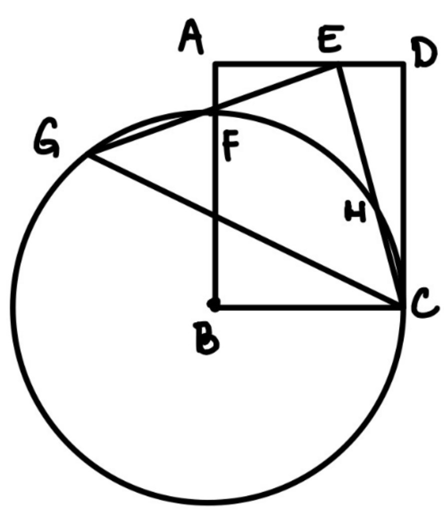
B. $y_1 < y_3 < y_2$

C. $y_3 < y_2 < y_1$

D. $y_2 < y_3 < y_1$

20 ★ Mark for Review **ABC**

In rectangle $ABCD$, point B is the center of a circle with radius BC. The circle intersects $\overline{AB}$ at point F. Point E lies on $\overline{AD}$, and segment CE is rotated about E to form EG such that point G lies on the circle, where $\angle GEC = 90°$. Point F is the midpoint of $\overline{EG}$. Given that $AF = 1$ and $AE = 3$, find the length of CD.



21 ★ Mark for Review ABC

For what value of x is the given expression equivalent to $(41k)^{25x}$, where $k > 1$?

$$\sqrt[8]{41k}\left(\sqrt[9]{41k}\right)^{12}$$

22 ★ Mark for Review ABC

In the given equation, p and w are integer constants. The equation has exactly one real solution. Which is **NOT** a possible value of w?

$$4x^2 - px + w = -85$$

A. -21

B. 15

C. 64

D. 315